Q585 re-aimed: Purbhoo's flock swap equals the crystal commutor (computed on 190 cells, $c^{\nu}_{\lambda\mu} \in \{2,3,4,5\}$, under two bijections)

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2026-11-12, PROVE session c2.

To: Robin. From: Clio. Provenance (PROTOCOL §2.3): sources commit `7303d1db5f52` (clio-vega/proofs, `code-1112c2-Q585/`); addendum §7: `3257a1d79e78` (`switch.py`); C instrument from `code-1112-Q585/` (PROVE c1). Source read first-hand: Purbhoo, arXiv:0705.1184, §§2–4 and §5.1 (text, and the figures on pp. 6 and 13 rendered at 400–900 dpi). **Grade: computed, not proved.**

Abstract

Purbhoo proves commutativity of the Littlewood–Richardson rule bijectively by migrating flocks of $(150°, 30°)$ rhombi through a mosaic [1, Cor. 3.4]. The result is a map $M : \mathrm{LR}(\lambda/\mu;\nu) \to \mathrm{LR}(\lambda/\nu;\mu)$, which he states is an involution [1, Prop. 5.4]. I built migration from the definitions, exactly and from scratch. I then compared M with the Henriques–Kamnitzer crystal commutor C on every cell with $c \geq 2$ in the range $\mathfrak{gl}_4$, $|\lambda|, |\mu| \leq 7$ ($c \geq 3$) or ≤ 5 ($c = 2$). **They agree on** 518/518 **elements in** 190/190 **cells, under each of two independent tableau↔highest-weight bijections.** The lex control separates from C in every cell. This is computed evidence and not a proof. The gap is stated in §6.

1 The question and the pre-registration

The brief (WAKE 2026-11-12 c2) asked whether Purbhoo's swap equals C on highest-weight elements. It pre-registered "I don't know; both outcomes informative" and named the falsifier as one $c \geq 2$ cell where the swap differs from C, under a bijection that passes its own round-trip control. The falsifier did not fire.

2 Definitions, as read ([1] §§2.3–3.3)

1. *Mosaic.* A mosaic is a tiling of the hexagon $A'AB'BC'C$ (angles 150° at A, B, C and 90° at A', B', C') by unit triangles, unit squares and $(150°, 30°)$ rhombi, with the rhombi packed into nests at A, B, C. Mosaics are in bijection with KT puzzles: delete the rhombi and flex the squares to 60° rhombi; the 0-edges and 1-edges rotate in opposite senses. The standard orientations are $(E_A, N_A) = (\vec{AA'}, \vec{AB'})$, $(E_B, N_B) = (\vec{BB'}, \vec{BC'})$, $(E_C, N_C) = (\vec{CC'}, \vec{CA'})$.

2. *Flock.* A flock is $(\zeta, (E, N), f)$, where ζ is a set of nest rhombi forming a transformed skew diagram and f is an LR tableau on ζ in the frame (E, N). The *standard order* compares f-values first and breaks ties by "west first".

3. *Migration* (§3.1). Turn the page so that the direction of migration points right. N/E migration processes the rhombi from the standard-order maximum downward, and S/W migration from the minimum upward. Each rhombus $\Diamond$ repeatedly finds the smallest hexagon containing it

that lies weakly right of its leftmost point. Figures (a) and (b) on p. 13 show both kinds as *rhombus + square + two triangles*. If the hexagon is 2-fold symmetric (a), rotate its tiling by 180°. If it is 3-fold symmetric (b), rotate by 120° in the sense that leaves an edge of ◊ horizontal or vertical. Stop when ◊ is packed in a new nest.

4. *The swap* (Cor. 3.4). Let f be an LR tableau on λ/μ with content ν. Form the canonical mosaic $(\varnothing, \gamma^\vee, \gamma)$ with $\gamma^\vee = \lambda$ in (E_B, N_B) (Lemma 2.6). Let F be the flock of f, and G the unique flock on the remaining rhombi, in orientation $(-E_B, -N_B)$. Migrate F from B to A, then G from B to A. Then G' is an LR tableau on λ/ν with content μ.

What the paper does not make constructive. Lemma 2.6 (the canonical mosaic) is stated with "proofs omitted"; I build it from the unique KT puzzle with that boundary and assert uniqueness. The proofs of Prop. 5.4 and of the jdt dictionary (Props. 5.1–5.2) are omitted. Cor. 3.3 is a *family* of swaps ("fix any orientations"), so "the" Purbhoo swap is well-defined only for Cor. 3.4, which is what is tested here. Text-level readings: Cor. 3.4 says η "corresponds to ν", but the remaining rhombi are μ, so I read it as μ; §2.2(i) says "east to west", but Fig. 9 increases west to east, and I follow Fig. 9.

3 The instrument

Exact geometry. A point is the KT lift $P = (P_0, P_1) \in \mathbb{Z}[\omega]^2 \cong \mathbb{Z}^4$, the sums of the 0-labelled and 1-labelled edge vectors along any path, as in Knutson–Tao's $\mathbb{R}^4$ surface ([1, §2.4]). It is drawn as $\phi(P) = e^{i\pi/12}P_0 + e^{-i\pi/12}P_1$. The map ϕ is injective, since $e^{i\pi/6} \notin \mathbb{Q}(\omega)$. Rotations by $60°k$ act componentwise and exactly, so every tile coordinate is an integer vector. Floats are used only for the left/right and horizontal tests. Of the four sign/corner conventions, *exactly one* makes every KT rhombus a square and every mosaic convex and area-exact (41/41 at $n = 4$). So the convention is forced, not fitted.

Corrections found by the instrument, in order. (i) My first low-resolution reading of Fig. (a) took it as three tiles. At 900 dpi it is four: both moves are rhombus + square + two triangles. (ii) The union's boundary has a straight-angle vertex ($30 + 90 + 60 = 180$), so "hexagon" means six *corners* on seven unit edges. (iii) In 956 steps two hexagons tie for minimal. Both lie weakly right of the leftmost point, which contradicts the paper's uniqueness parenthesis. I break the tie by the pattern visible in both figures: *the hexagon contains the tile across the horizontal edge of ◊ that starts at its leftmost vertex.* This is an interpretation, and it is flagged. It is load-bearing: the lesion that takes the other candidate gets stuck on 63/142 mosaics at $n = 5$. With the rule, all ties resolve (956/956).

Validation against the paper's own theorems , before Cor. 3.4 was touched. On all mosaics with $n = 4, 5$ ($27 + 112$), for all 8 orientation/direction combinations of Table 1, the unique flock on α migrates *every* time. It lands in the nest the text predicts (for example $(-E_A, -N_A)$ south $\to B$, as in Cor. 3.2), and the result is an LR tableau (Thm. 1) in 100% of cases. Cor. 3.2 is injective within each boundary: 142 mosaics, 0 collisions.

4 Transport: two bijections

C acts on $\mathrm{hw}(B(\lambda)\otimes B(\mu))_\nu$, realised as $b \in \mathrm{SSYT}(\lambda)$ with $\mathrm{read}(b)\,\mathrm{read}(u_\mu)$ Yamanouchi. It is computed by two instruments from c1 (crystal operators; RSK plus evacuation), which agree on 1308/1308 elements.

- β_{CM}: the content-matrix transpose. Row r of ν/μ holds $m_{k,r}$ copies of k, where $m_{k,r} = \#\{r{+}1 \text{ in row } k \text{ of } b\}$. This gives $\mathrm{hw} \to \mathrm{LR}(\nu/\mu;\lambda)$.
- β_Q: the recording tableau. Write $w = \mathrm{read}(b)\,w'$, where $w' = \mathrm{RSK}^{-1}(u_\mu, U_\mu)$ and U_μ is row-superstandard, so that $w' \equiv u_\mu$ (Knuth) with a monotone destandardisation. The entries of $Q(w)$ above $|\lambda|$ form a standard skew tableau on ν/λ that rectifies to U_μ. Relabel each entry by its row in U_μ. This gives $\mathrm{hw} \to \mathrm{LR}(\nu/\lambda;\mu)$. It is valid and injective on 1308/1308. Two naïve versions failed their own round-trip control and were discarded: recording the inserted letter gives anti-LR tableaux (990/1308 bad), and a row-reading relabel is non-monotone (326/1308 bad).

The two bijections land in *different* sets: β_{CM} feeds M an input on ν/μ, and β_Q one on ν/λ. So they test M on different inputs.

5 Result

Proposition 1 (computed)**.** *For every $\mathfrak{gl}_4$ cell with $c^\nu_{\lambda\mu} \geq 2$ and $|\lambda|, |\mu| \leq 5$, and every cell with $c^\nu_{\lambda\mu} \geq 3$ and $|\lambda|, |\mu| \leq 7$, and for $\beta \in \{\beta_Q, \beta_{\mathrm{CM}}\}$,*

$$M \circ \beta \;=\; \beta \circ C \quad \textit{on } \mathrm{hw}(B(\lambda)\otimes B(\mu))_\nu.$$

c	cells	elements	agree (β_Q)	agree (β_{CM})	lex $\neq C$
2	67	134	134	134	67/67
3	109	327	327	327	109/109
4	13	52	52	52	13/13
5	1	5	5	5	1/1
all	190	518	518	518	190/190

In every case the output of M is an LR tableau of the predicted shape (Thm. 1 at scale), and $M \circ M = \mathrm{id}$ (Prop. 5.4 confirmed, 0 failures under either bijection).

How many bits this is. In a $c = 2$ cell there are only two bijections, and the c1 controls showed that lex $\neq C$ and reverse-lex $= C$ there. So on the 67 $c = 2$ cells, "$M \neq$ lex" and "$M = C$" are *the same bit* (the c1 lesson: a cheap control can be the target in disguise). Those cells carry 67 bits, not more. The $c \geq 3$ cells are the real test. A $c = 3$ cell has 6 bijections and a $c = 4$ cell has 24, and M hits C's in all 123. There, lex differs from C in 123/123 cells, but reverse-lex differs in only 27/123. So reverse-lex remains a *near-blind* control, and the separation from it rests on those 27 cells.

A lesion that must fire. Replace G's orientation $(-E_B, -N_B)$ by (E_B, N_B) and migrate it east. This is a legal migration but not Purbhoo's recipe. It differs from C on 67/67 $c = 2$ cells. It is a crude lesion: 111 of its outputs are not even LR in that frame.

6 What is and is not established

- **Computed:** Proposition 1, with the flagged tie-break rule as part of the definition of migration. The verdict does not depend on which of the two bijections is used.
- **Not proved.** A proof would need two steps. (1) A proof that Cor. 3.4's migrations are a specific jdt composite. Purbhoo's §5.1 asserts that migration corresponds to jdt (Props. 5.1, 5.2, Cor. 5.3), but omits the proofs, and the B→A migrations used in Cor. 3.4 are not covered by the stated B→C case. (2) An identification of that composite with $C = \xi_{B(\mu)\otimes B(\lambda)} \circ (\xi \otimes \xi) \circ \mathrm{flip}$. The natural candidate is a tableau-switching description of C. I recall such an identification in the literature (Benkart–Sottile–Stroomer switching, and an equivalence result for LR symmetry maps), but *from memory only, not read.* So it is not cited here and nothing rests on it.
- **Consequence for Q583/PV05** (*hunch-grade*, inherited framing): the "migration train" at the Schur point induces the coboundary commutor on highest-weight elements. On this axis PV05 is not a rival *involution*: same map, different construction. That says nothing about the YBE/R-matrix train of c1, which remains ill-posed at the instrument.

7 Addendum (same session, re-entry): the gap splits through tableau switching

The intermediate map. For an LR tableau S on ν/κ of content ρ, define $\Sigma(S)$ as follows. Place u_κ on κ. Take the cells of κ in decreasing standard order of u_κ: larger entry first, equal entries right to left. Slide S inward into each cell by ordinary jeu de taquin (the hole takes the smaller of its right and next-row neighbours; on a tie it takes the next-row one). Label the outer cell vacated by that slide with the u_κ-entry of the starting cell. The labels form $\Sigma(S)$ on ν/ρ. This is switching of the pair (u_κ, S) in the sense of Benkart–Sottile–Stroomer. The name is recalled, not read, and nothing below rests on any property of switching from the literature: every property used is checked by computer.

Proposition 2 (computed; same census as Prop. 1: 190 cells, 518 elements)**.** *(H1)* $M(T) = \Sigma(T)$ *for every* $T = \beta_{\mathrm{CM}}(b) \in \mathrm{LR}(\nu/\mu; \lambda)$*:* 518/518*.*

(H2) $\Sigma \circ \beta = \beta \circ C$ *for* both $\beta = \beta_{\mathrm{CM}}$ *(*518/518*) and* $\beta = \beta_Q$ *(*518/518*).*
Moreover, $\Sigma(T)$ *is always LR of content* μ *(*518/518*), and* $\Sigma \circ \Sigma = \mathrm{id}$ *(*518/518*).*

Controls. Σ differs from lex in 190/190 cells. Two lesions of Σ, namely breaking ties toward the right neighbour and processing κ column-first, together differ from Σ in 190/190 cells. So Σ is not an order-insensitive artefact.

What this buys. H1 and H2 together imply Prop. 1. Each is a separate claim, and so the single open gap becomes two gaps of different kinds:

- **H1 is the Purbhoo-specific half.** It says that the two B→A migrations of Cor. 3.4 together *are* switching against u_λ. A proof would go through Purbhoo's jdt dictionary (§5.1). That dictionary has its proofs omitted and is stated only for B→C. So H1 is a *proof* task about mosaics.

- **H2 is Purbhoo-free.** It identifies switching with the HK commutor on highest-weight elements. I hold this in the literature only at agent-summary grade (e.g. `0906.0077`), so H2 is a *reading* task. It is not cited here.

Neither half is proved in this note.

8 Verification artifacts

`proofs/code-1112c2-Q585/`: `mosaic.py` (puzzles, lift, mosaics), `migrate.py` (moves, journeys), `flock.py` (flocks, frames, Cor. 3.4), `geotest.py` (convention 1 of 4), `cor32.py`/`cor32b.py` (Table-1 validation, injectivity, tie-break lesion), `betaq.py`, `compare.py`, `compare2.py` (the result), `ctl3.py` (controls), `lesion34.py`, `switch.py` (Prop. 2), `results-*.txt`, `DEFS.md` (the definition summary written before coding).

References

[1] K. Purbhoo, *Puzzles, tableaux, and mosaics*, arXiv:0705.1184 (J. Algebraic Combin. 2008), §§2–5.

[2] A. Henriques, J. Kamnitzer, *Crystals and coboundary categories*, arXiv:math/0406478 (the commutor formula used, as implemented in c1; formula quoted via `0707.2248` l.855 in c1).